Kinetic Theory of Gases: Maxwell's Speed Distribution

This distribution describes how the different speeds are distributed among various group of molecules.

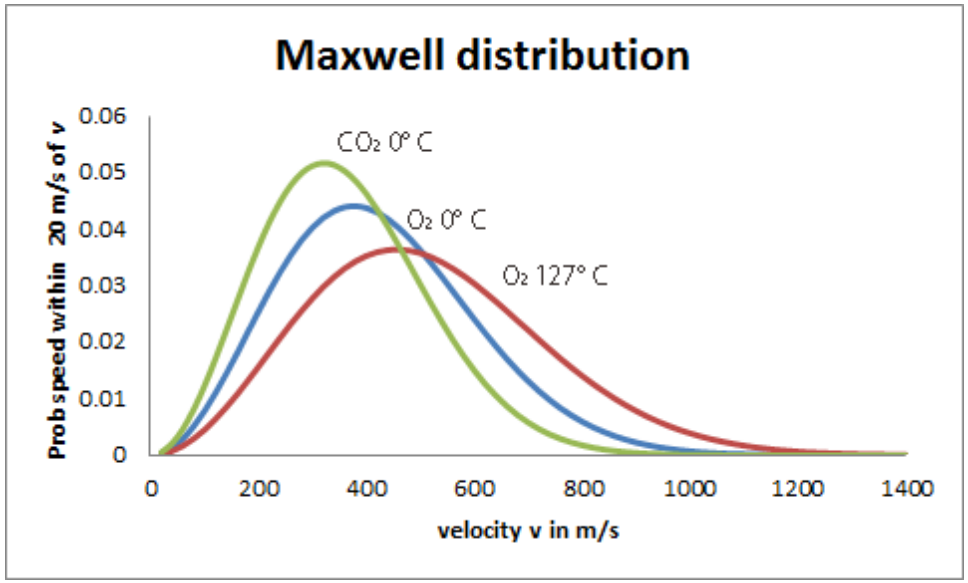
$$F(c) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} \cdot c^2 \cdot e^{-\frac{Mc^2}{2RT}}$$

$F(c) \rightarrow$ Maxwell speed distribution function

where $F(c) = \frac{1}{n} \frac{dn_c}{dc}$

n = no of molecules per unit volume

dn_c = " " " within the speed range c to $(c + dc)$

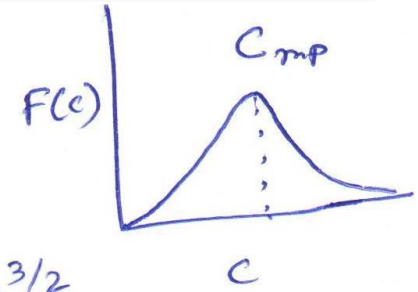


- ✓ **Salient features of Maxwell's distribution curve**
- ✓ **Effect of temp. on the distribution**
- ✓ **Effect of molar mass on distribution**
- ✓ **Kinetic energy distribution from Maxwell's speed distribution**

- ✓ **If $T_2 > T_1$, then $c_{mp}(T_2) > c_{mp}(T_1)$, but $F(c_{mp})$ at $T_2 < F(c_{mp})$ at T_1**
- ✓ **When T ↑, the curve becomes more and more flat.**
  When $T \rightarrow \infty$, the curve lies on the c-axis indicating that the speed distribution is totally uniform.
  All the molecules have same speed and in fact distribution is lost.
- ✓ **If T ↓, the curve becomes narrow. When $T \rightarrow$ 0 K, the curve lies on the F(c) – axis.**
  All the molecules have speed zero and so again distribution is lost.

Maxwell's Speed Distribution: Derivation of most probable speed

$$F(c) = \frac{1}{n}\frac{dn_c}{dc} = 4\pi\left(\frac{m}{2\pi RT}\right)^{3/2} \cdot c^2 \cdot e^{-\frac{mc^2}{2RT}}$$



Differentiating F(c) w.r.t c

$$\frac{d}{dc}[F(c)] = A\left[2c\, e^{-\frac{mc^2}{2RT}} - e^{-\frac{mc^2}{2RT}} \cdot \frac{2mc}{2RT} \cdot c^2\right], \text{ where } A = 4\pi\left(\frac{m}{2\pi RT}\right)^{3/2}$$

At maxima, $\frac{d}{dc}[F(c)] = 0$ Thus, $Ac\, e^{-\frac{mc^2}{2RT}}\left[2 - \frac{mc^2}{RT}\right] = 0$

Now, since $A \neq 0$

Either i) $c = 0$

or ii) $e^{-\frac{mc^2}{2RT}} = 0$ or $c \Rightarrow \infty$

or iii) $2 - \frac{mc^2}{RT} = 0$

But, case (i) and (ii) correspond to the minima situation as evident from distribution curve.

Thus, for the maxima, $2 - \frac{mc_{mp}^2}{RT} = 0$

$$\Rightarrow c_{mp} = \sqrt{\frac{2RT}{m}}$$

Maxwell's Speed Distribution: Applications

Average of any molecular property which is function of speed can be derived using Maxwell's speed distribution;

$$q = f(c)$$

$$\bar{q} = \frac{1}{n}\int_0^\infty q \, dn_c$$

q: molecular property which is function of speed.

$$F(c) = \frac{1}{n}\frac{dn_c}{dc} = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2} \cdot c^2 \cdot e^{-\frac{mc^2}{2kT}}$$

① Derivation of average speed ($\bar{c}$) :-

$$\bar{c} = \frac{1}{n}\int_0^\infty c \, dn_c$$

$$= \frac{1}{n}\int_0^\infty c \cdot 4\pi n \left(\frac{m}{2\pi kT}\right)^{3/2} \cdot c^2 \cdot e^{-\frac{mc^2}{2kT}} \cdot dc$$

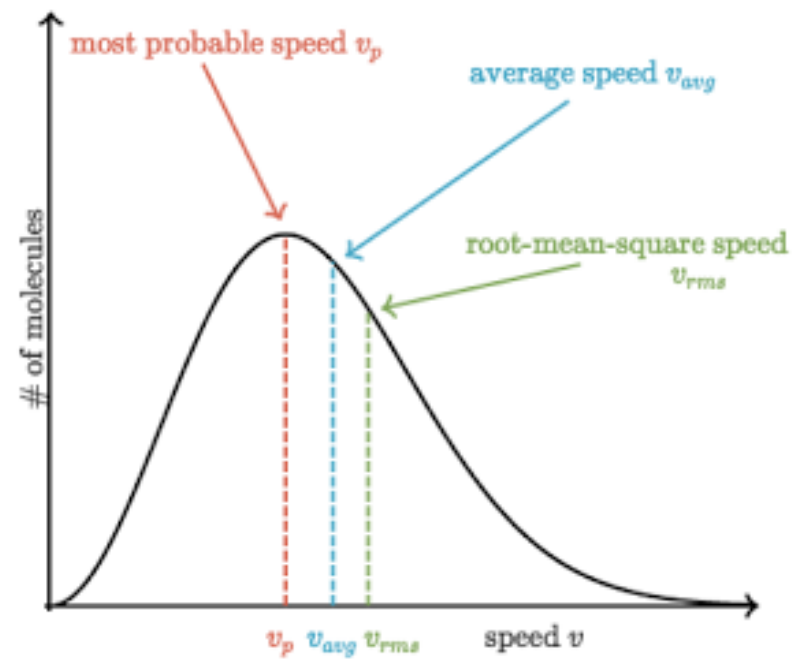
$$\bar{c} = \sqrt{\frac{8kT}{m\pi}}$$

② Derivation of root mean square speed :-

$$\overline{c^2} = \frac{1}{n}\int_0^\infty c^2 \, dn_c$$

$$\vdots$$

$$\sqrt{\overline{c^2}} = \sqrt{\frac{3kT}{m}}$$



$$C_{mp} = \sqrt{\frac{2kT}{m}}$$

$$\bar{c} = \sqrt{\frac{8kT}{m\pi}}$$

$$\sqrt{\overline{c^2}} = \sqrt{\frac{3kT}{m}}$$

For Neon gas at room temperature (293 K);

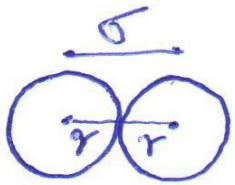
C_{mp} = 491 m/s ; C_{avg} = 554 m/s ; C_{rms} = 602 m/s

Collision Number and Collision Frequency

Collision number:- Number of collisions made by gas molecules in unit time

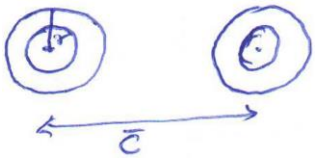
Collision frequency:- Number of collisions occurring between gas molecules in unit time and in unit volume

As the molecules are in random motion, so they collide with themselves also. When the molecules collide, the center of two molecules can not approach beyond a certain distance, called as distance of closest approach or collision diameter (σ) of the molecules, where $\sigma = 2r$.



Therefore, each molecule can be considered to be surrounded by an imaginary sphere of radius 2r such that if any other molecules come within this spherical region, it will make a collision with the molecule under consideration. This sphere is called "sphere of influence".

Let, the molecule travel $\bar{c}$ distance in 1 sec



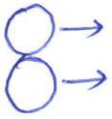
$\therefore$ Volume swept by the imaginary sphere in 1 sec = $\pi\sigma^2 \bar{c}$

If n be the number of molecules per unit volume, then number of molecules within the swept volume or in other words number of collision of molecules will make with other molecules in 1 sec = $\pi\sigma^2 \bar{c}\, n$

Since all the molecules are mobile, therefore in the above calculation relative velocity must be considered.

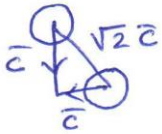
Two molecules may collide at different possible angles but two extreme cases are collision at 180° (head on collision) and collision at 0° (grazing collision).

Head on collision;
Angle = 180°



Grazing collision;
Angle = 0°

Average collision angle = 90°



Collision Number and Collision Frequency

No. of collisions in 1 sec = $\pi\sigma^2 \bar{c}\, n$

Since the relative velocity is $\sqrt{2}\,\bar{c}$, therefore, no. of collisions one molecule makes in 1 sec $= \pi\sigma^2\sqrt{2}\,\bar{c}\cdot n$

$$Z_1 = \sqrt{2}\,\pi\sigma^2\bar{c}\,n$$

$\therefore$ Collision frequency should be equals to $\sqrt{2}\,\pi\sigma^2\bar{c}\,n\cdot n$

To avoid double counting the above number should be devided by 2. Hence, collision frequency betwy. similar molecules,

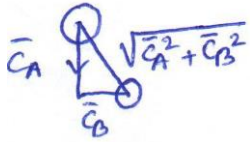
$$Z_{AA} = \frac{1}{2}\cdot\sqrt{2}\,\pi\sigma^2\bar{c}\,n^2$$

$$= \frac{1}{\sqrt{2}}\,\pi\sigma^2\bar{c}\,n^2$$

(per unit time per unit volume)

Collision frequency between dissimilar/different molecules:-

In this case avg. collision diameter $= \dfrac{\sigma_A + \sigma_B}{2}$



Relative resultant velocity $= \sqrt{\bar{c}_A^2 + \bar{c}_B^2}$

$\bar{c}_A$: Avg velocity of A molecule

$\bar{c}_B$: " " " B "

$$\bar{c}_A = \sqrt{\frac{8kT}{\pi m_A}}$$

$$\bar{c}_B = \sqrt{\frac{8kT}{\pi m_B}}$$

$$\sqrt{\bar{c}_A^2 + \bar{c}_B^2} = \sqrt{\frac{8kT}{\pi}\left(\frac{1}{m_A}+\frac{1}{m_B}\right)} = \sqrt{\frac{8kT}{\pi\mu}} \quad \left[\mu = \frac{m_A m_B}{m_A + m_B} = \text{reduced mass}\right]$$

So, collision frequency $= \pi\left(\dfrac{\sigma_A+\sigma_B}{2}\right)^2\sqrt{\dfrac{8kT}{\pi\mu}}\, n_A n_B$

n_A = no. of A molecules

n_B = no. of B "

Thus, $Z_{AB} = (\sigma_A+\sigma_B)^2\sqrt{\dfrac{\pi kT}{2\mu}}\, n_A n_B$

Since collision between dissimilar molecules are considered, no double counting takes place.

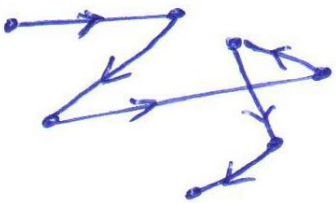
Mean free path (l)

Mean free path (l) :-

- ✓ **It is the average distance between two successive collisions of a molecule**
- ✓ **The free path of a molecule can range from 0 to ∝**
- ✓ **A molecule can collide with another molecule just immediately after the start of its motion or it can move a long distance without suffering any collision**

$l = \frac{\bar{c}}{Z_1}$; where Z_1 is the number of collisions made by a molecule in unit time when it travels $\bar{c}$ distance.

$= \frac{\bar{c}}{\sqrt{2}\pi\sigma^2\bar{c}n} = \frac{1}{\sqrt{2}\pi\sigma^2 n}$; n = no. of molecules per unit volume.



Effect of Pressure (P) and Temperature (T) on mean free path :-

Mean free path decreases with increasing P and increases with increasing T. $l \downarrow P \uparrow$; $l \uparrow T \uparrow$

Mathematical proof:- $l = \frac{1}{\sqrt{2}\pi\sigma^2 n}$ — (1)

we know for n' mole of ideal gas; $PV = n'RT = \frac{N}{N_A}RT$

N = Total no. of molecules

N_A = Avogadro's no.

$\Rightarrow P = \frac{N}{VN_A}RT = \frac{n}{N_A}RT$

n = no. of molecules per unit volume

Thus, $n = \frac{PN_A}{RT}$

Replacing n in eqn. (1); $l = \frac{1}{\sqrt{2}\pi\sigma^2}\frac{RT}{PN_A}$

Thus, $l \propto T$ and $l \propto \frac{1}{P}$

Again, $PV = n'RT$

$= \frac{w}{M}RT$ [w = mass of the gas, M = Molar mass]

$\Rightarrow P = \frac{w}{VM}RT = \frac{\rho}{M}RT$; Thus, $l = \frac{1}{\sqrt{2}\pi\sigma^2} \times \frac{M}{\rho N_A}$

$\Rightarrow \frac{RT}{P} = \frac{M}{\rho}$

$\Rightarrow l \propto \frac{1}{\rho}$

As $\rho \uparrow$, the molecule come closer to each other, thus increasing the number of collision, decreasing mean free path.

Mean free path (I)

For an isochoric system, the pressure also changes with temperature (directly proportional). So, how can we measure the dependence of the mean free path on these two parameters?

Isochoric process: A thermodynamic process in which the volume of the closed system remains constant (**V = const**)

$$l = \frac{1}{\sqrt{2}\,\pi\sigma^2 n} \quad \text{— (1)}$$

Since *n* is constant in Eq (1)

↓

l = constant

The mean-free path in the isochoric process is independent of temperature

we know for n' mole of ideal gas; $PV = n'RT = \frac{N}{N_A}RT$

N = Total no. of molecules

N_A = Avogadro's no.

$$\Rightarrow P = \frac{N}{VN_A}RT = \frac{n}{N_A}RT$$

n = no. of molecules per unit volume

Thus, $n = \frac{PN_A}{RT}$

Replacing n in eqn. (1); $l = \frac{1}{\sqrt{2}\,\pi\sigma^2}\frac{RT}{PN_A}$